An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Kelani Riverine Communities in Sri Lanka

R26-IT-162

SLIIT FACULTY OF COMPUTING

An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Kelani Riverine Communities in Sri Lanka

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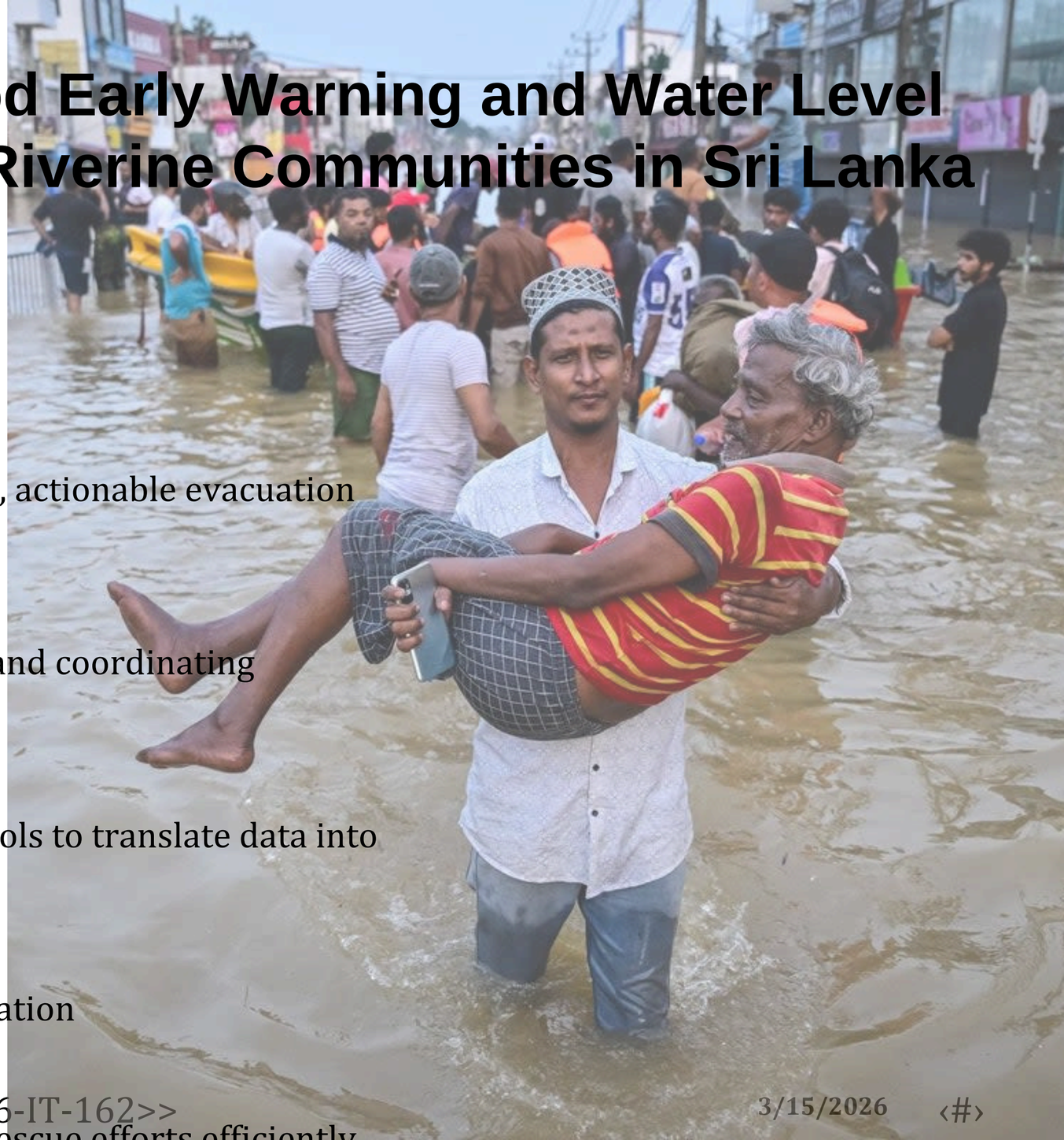
USERS AND STAKEHOLDERS



Riverine Communities (Primary Users)
Households along Kelani River who need personalized, actionable evacuation information in their language

Disaster Management Centre (DMC)
Government agency responsible for issuing warnings and coordinating evacuations

Irrigation Department
Provides authoritative river level data; needs better tools to translate data into community intelligence

Department of Meteorology
Provides rainfall forecasts; stakeholder for data integration

Emergency Responders
Need dynamic evacuation priority lists to coordinate rescue efforts efficiently

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IMPACT OF THE PROBLEM

Impact Area	Magnitude
Human Cost	646 fatalities, 173 missing (Cyclone Ditwah 2025)
Economic Cost	US$4.1 billion total damages (~4% of Sri Lanka's GDP)
Livelihood Impact	93% of affected communities reported severe livelihood impacts
Social Exclusion	Tamil-speaking communities excluded from critical warnings

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EXISTING SOLUTIONS AND THEIR LIMITATIONS

Solution Type	Examples	Limitations
Institutional Early Warning	DMC SMS/TV alerts, IWMI AWARE platform	Macro-level only, no personalization, language barriers
Academic Research Prototypes	SLIIT LoRa-ML, Moratuwa Bi-LSTM models	Siloed, validation-stage only, not operational for communities
Satellite-Based Systems	Wuhan University SAR monitoring	Post-disaster only, no predictive capability, low resolution

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THE PROPOSED SOLUTION

An integrated, intelligent IoT-based flood early warning system that combines:

- Hyperlocal IoT Rainfall Monitoring - Solar-powered sensor nodes along Kelani Basin
- Hybrid Deep Learning Prediction - Bi-LSTM + Time Series Regression (0.95 validated accuracy)
- Geospatial Risk Personalization - Household-level Dynamic Flood Risk Scores
- Multilingual Community Alerts - Sinhala/Tamil/English with location-specific guidance

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HOW IT OVERCOMES LIMITATIONS

Existing Limitation	How Our Solution Addresses It
Generic, macro-level warnings	Household-specific risk scores based on distance from river and rate of rise
Language exclusion	Automated alerts in Sinhala, Tamil, and English
No real-time local data	IoT rain gauge network at higher resolution than Met Dept stations
Fragmented sensing/prediction/communication	Unified framework integrating all components
No operational deployment	Built on validated research (Moratuwa 0.95 accuracy for Kelani) and authoritative government data

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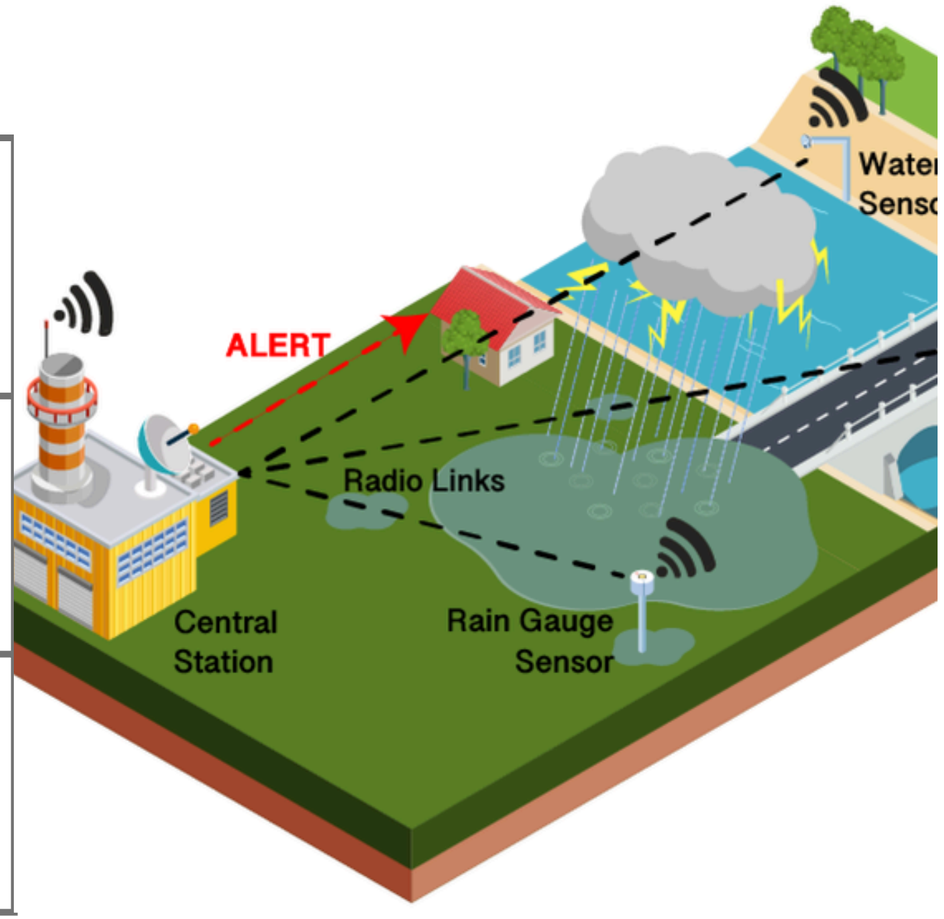
INDIVIDUAL STUDENT COMPONENTS

Student	Component	Key Contribution
Rasara S.H.K (IT21313998)	IoT-Based Hyperlocal Rainfall Monitoring + Hybrid Deep Learning Prediction	Real-time rainfall data + 1-3 day water level forecasts using Bi-LSTM
U. H. K Sankalana (IT22236814)	Distance-Based Dynamic Flood Risk Personalization + Multilingual Alerting	Household risk scores + multilingual push notifications



IT21313998 Rasara S.H.K

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IoT-Based Hyperlocal Rainfall Monitoring and Hybrid Deep Learning Flood Prediction

CONTRIBUTION TO OVERALL SOLUTION

My component provides the intelligence backbone of the system:

Input	Process	Output to Partner
Real-time rainfall from deployed IoT sensors	Bi-LSTM + Time Series Regression hybrid model	24/48/72-hour water level forecasts for Kelani River stations
Historical rainfall & river data (10+ years)	Model training and validation	Real-time rainfall intensity data
Irrigation Department water level data	Prediction API deployment	

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NOVELTY

Novelty 1: Hyperlocal IoT Rainfall Network for Kelani Basin - First deployment of low-cost, solar-powered IoT rain gauge network specifically designed for Kelani River basin

Novelty 2: Rainfall-Integrated Hybrid Deep Learning Architecture for Kelani River - First Bi-LSTM + Time Series Regression hybrid model for Kelani River that explicitly incorporates real-time IoT rainfall data as a dynamic exogenous input variable

IoT-Based Hyperlocal Rainfall Monitoring and Hybrid Deep Learning Flood Prediction

Research Problem

Current flood prediction models for Kelani River rely solely on historical water level sequences without incorporating real-time, hyperlocal rainfall data, reducing accuracy during rapidly developing flood events.

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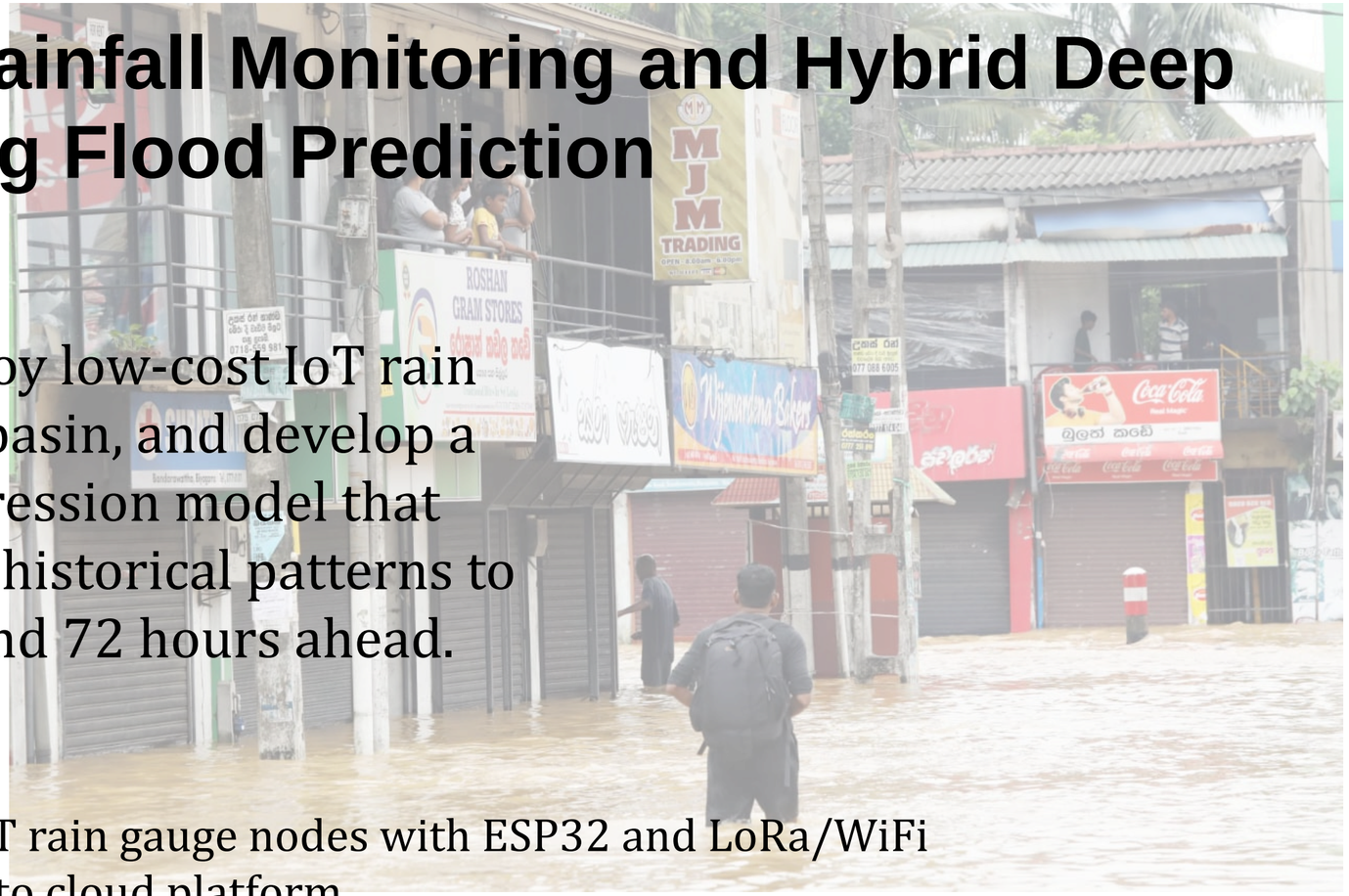

RESEARCH OBJECTIVE AND SUB-OBJECTIVES

Main Objective: To design and deploy low-cost IoT rain gauge nodes along the Kelani River basin, and develop a hybrid Bi-LSTM + Time Series Regression model that integrates real-time rainfall data with historical patterns to forecast river water levels 24, 48, and 72 hours ahead.

Sub-Objectives:

- Design and assemble 3-5 solar-powered IoT rain gauge nodes with ESP32 and LoRa/WiFi
- Establish real-time data ingestion pipeline to cloud platform
- Collect and preprocess 10+ years of historical hydrological data (Irrigation Dept + CRAN "Ditwah" R package)
- Implement and train Bi-LSTM + Time Series Regression hybrid model using TensorFlow/Keras
- Integrate real-time IoT rainfall as exogenous input variable
- Validate model against Cyclone Ditwah event data and compare with ARIMA/SVR/LSTM baselines
- Deploy prediction API for integration with risk scoring engine

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APPLICATION OF DOMAIN KNOWLEDGE

Knowledge Domain	Application in Methodology
Embedded Systems	ESP32 programming, sensor interfacing, LoRa communication protocols
IoT Architecture	Solar power management, deep sleep modes, data transmission
Deep Learning	Bi-LSTM networks, Time Series Regression, TensorFlow/Keras implementation
Time Series Analysis	Sequence modeling, feature engineering, exogenous variable integration
Hydrological Modeling	Understanding river dynamics, rainfall-runoff relationships

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LATEST TECHNOLOGIES UTILIZED

Technology	Purpose
ESP32	Low-power microcontroller for sensor nodes
LoRaWAN	Long-range, low-power data transmission for rural areas
TensorFlow/Keras	Deep learning model implementation and training
Bi-LSTM + Time Series Regression	State-of-the-art hybrid architecture for flood forecasting (validated 80-100% accuracy for Sri Lankan basins)
REST API	Model deployment for integration with partner component
Pandas/NumPy	Data preprocessing and manipulation

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VALIDATION AND EVALUATION

Validation Method	Description
Model Accuracy	Compare predictions against actual water levels from Irrigation Department using MAE, RMSE, R^2
Benchmark Comparison	Compare against ARIMA, SVR, and standard LSTM baselines
Event Validation	Test against November 2025 Cyclone Ditwah event data
Sensor Calibration	Calibrate IoT rain gauges against Department of Meteorology reference stations
Cross-Validation	Time-series cross-validation on 10+ years of historical data

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DATA AVAILABILITY, SECURITY AND ETHICAL CLEARANCE

Aspect	Details
Data Sources	• Irrigation Department: Historical & real-time river levels (public API) • Department of Meteorology: Historical & forecasted rainfall • CRAN "Ditwah" R package: Kelani Basin datasets
Data Security	• Encrypted data transmission from sensors to cloud • Secure API endpoints with authentication • Regular data backups
Ethical Clearance	• No personal data collected • All data from public/government sources with permission • Will acknowledge all data sources in publications

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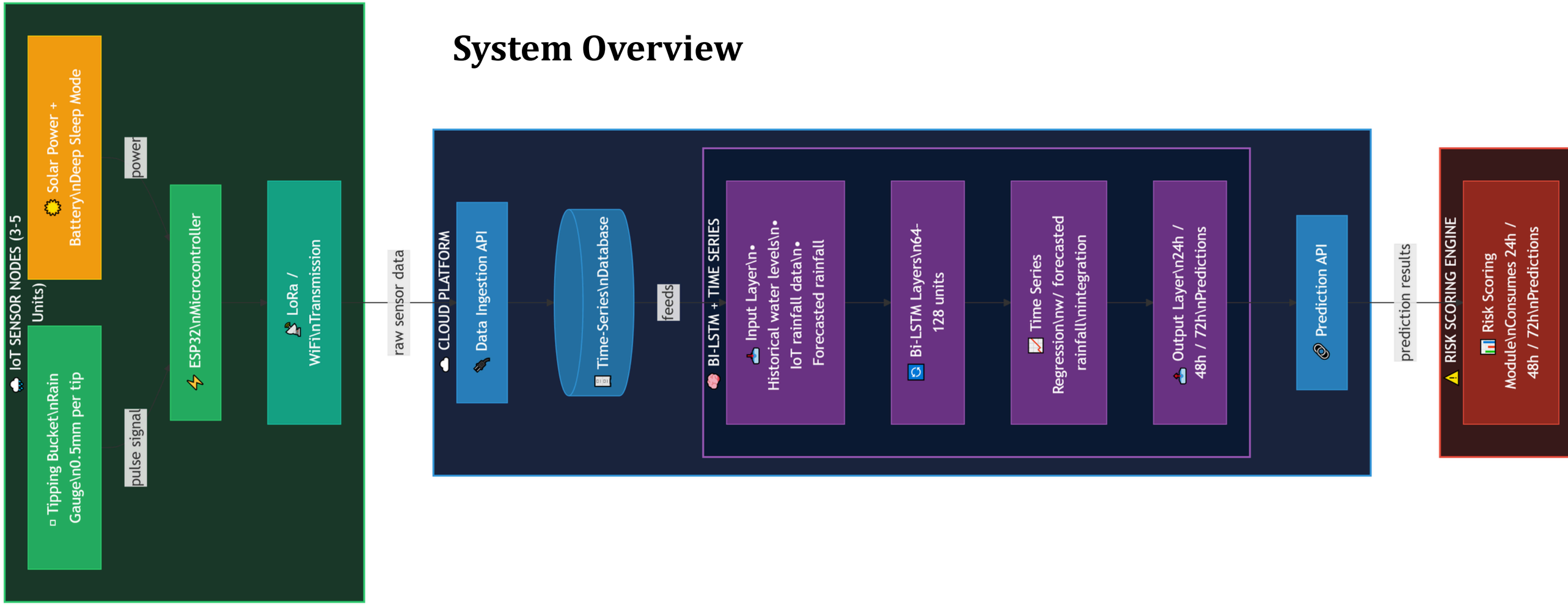
System Overview



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DEPENDENCIES AND COMMON COMPONENTS

Dependency	Type	Purpose
Irrigation Department API	External	Real-time river level data (shared with Student B)
Department of Meteorology API	External	Forecasted rainfall data
CRAN "Ditwah" R package	External	Historical Kelani Basin datasets
Cloud Database	Common	Shared data storage for both students
Prediction API	My Output	Consumed by Student B's risk engine
IoT Rainfall Data	My Output	Real-time rainfall intensity for risk scoring

IT22236814 U. H. K Sankalana

Distance-Based Dynamic Flood Risk Personalization and Multilingual Community Alerting

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CONTRIBUTION TO OVERALL SOLUTION

My component delivers the community-facing intelligence of the system:

Input	Process	Output
• Real-time river data (Irrigation Dept API) • 24/48/72h forecasts (from Student A) • IoT rainfall data (from Student A) • Household GPS coordinates	Dynamic Flood Risk Score (DFRS) algorithm + multilingual alert generation	• Personalized push notifications • Automated voice calls/SMS • Evacuation priority lists • Public web dashboard

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NOVELTY

- **Household-Level Dynamic Flood Risk Scoring Using Authoritative Government Data -** First system to translate Irrigation Department's 10-minute river data into personalized, household-specific evacuation intelligence

- **Linguistically-Inclusive Automated Alert System for Sri Lankan Flood Warnings** - First fully automated, multilingual (Sinhala/Tamil/English) flood alert system with location-specific evacuation guidance

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Research Problem

Current flood warnings in Sri Lanka are generic, delivered primarily in Sinhala, and lack household-specific information about evacuation urgency, leaving communities without actionable intelligence and excluding Tamil-speaking populations from life-saving information.

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RESEARCH OBJECTIVE AND SUB-OBJECTIVES

Main Objective: To develop a geospatial risk scoring engine that computes household-specific flood risk using real-time river data from Irrigation Department and rainfall data from Student A's IoT network, and deliver personalized multilingual alerts to at-risk communities.

Sub-Objectives:

- Implement API client for Irrigation Department 10-minute river data (Nagalagam Street, Hanwella, Kithulgala)
- Build geospatial database of registered household locations with GPS coordinates
- Develop Dynamic Flood Risk Score (DFRS) algorithm incorporating hydrological and geospatial factors
- Validate DFRS against historical flood inundation extents (DMC/IOM data)
- Design and develop React Native mobile app with push notifications
- Implement multilingual content delivery (Sinhala/Tamil/English)
- Create web dashboard for DMC and emergency responders
- Generate dynamic evacuation priority tiers

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APPLICATION OF DOMAIN KNOWLEDGE

Knowledge Domain	Application in Methodology
Geospatial Analysis	QGIS, GeoPandas for distance calculation, elevation modeling
Full-Stack Development	React Native (mobile), Flask/Django (web), REST APIs
Risk Modeling	DFRS algorithm design, multi-factor scoring, threshold definition
Multilingual Computing	Unicode support, language-specific content delivery
Human-Computer Interaction	Alert message design, user experience for vulnerable populations

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LATEST TECHNOLOGIES UTILIZED

Technology	Purpose
React Native	Cross-platform mobile application development
Flask/Django	Web dashboard for DMC and emergency responders
GeoPandas / QGIS	Geospatial analysis for distance and elevation calculations
Digital Elevation Models	Elevation above river level computation
Firebase Cloud Messaging	Push notification delivery
Twilio API / Local SMS Gateway	Automated voice calls and SMS in local languages
Unicode (Sinhala/Tamil)	Multilingual content rendering

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VALIDATION AND EVALUATION

Validation Method	Description
DFRS Validation	Compare risk scores against historical flood inundation extents from DMC and IOM/DTM data
Alert Accuracy	Verify alerts triggered correctly against official DMC thresholds
Language Verification	Native speaker review of Sinhala/Tamil/English alert content
User Testing	Small-scale focus group with Kelani River community members to assess alert clarity and usefulness
Performance Testing	Alert delivery latency, system scalability

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DATA AVAILABILITY, SECURITY AND ETHICAL CLEARANCE

Aspect	Details
Data Sources	• Irrigation Department: 10-minute river levels (public API) • Student A: IoT rainfall + 24/48/72h forecasts • DMC: Historical flood inundation maps • Household GPS: Collected from registered users with consent
Data Security	• Encrypted household location data • Secure user authentication • Role-based access control for DMC dashboard
Ethical Clearance	• Informed consent for GPS data collection • Data anonymized in publications • Compliance with Sri Lanka data protection guidelines • University ethics approval to be obtained

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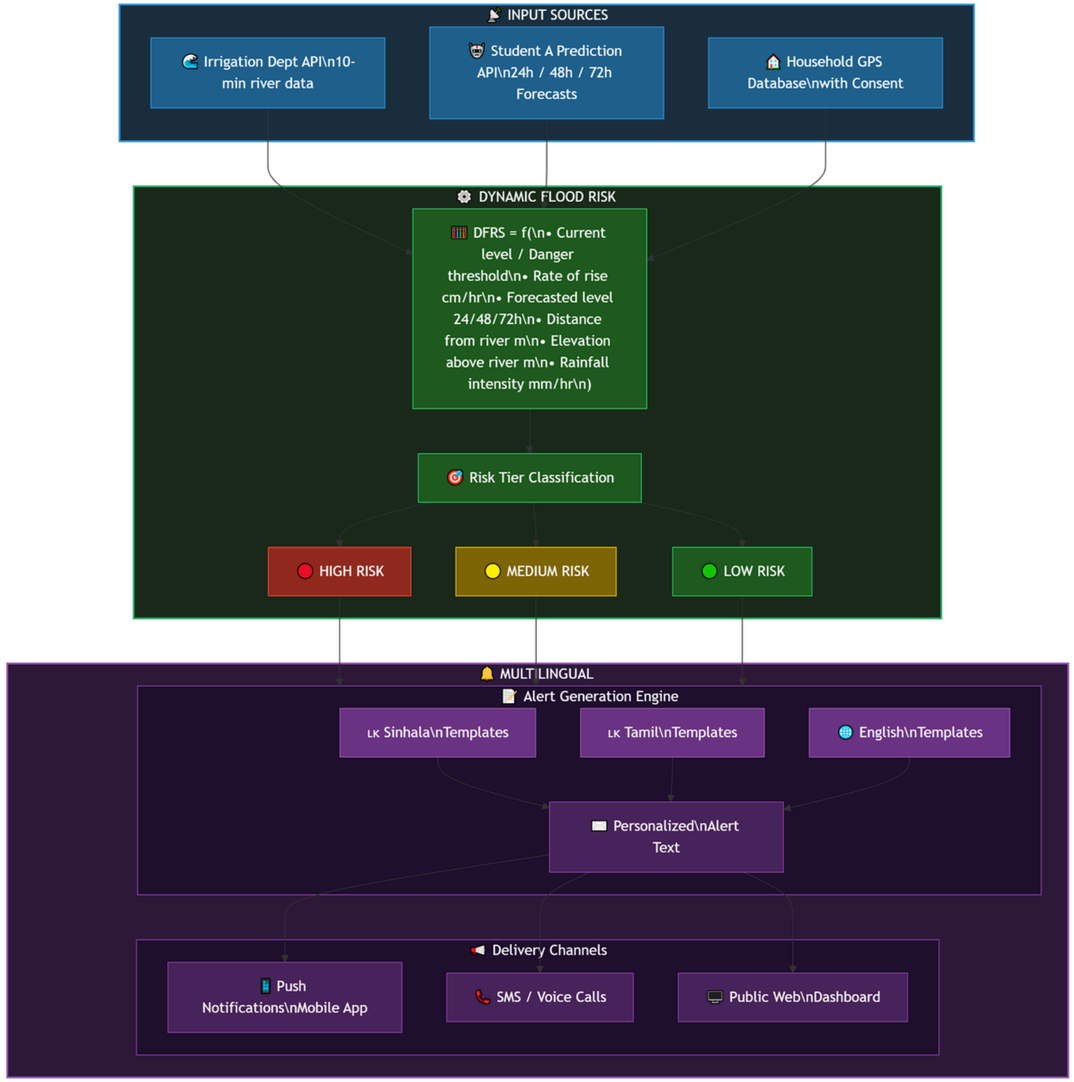
System Overview



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DEPENDENCIES AND COMMON COMPONENTS

Dependency	Type	Purpose
Irrigation Department API	External	Real-time river level data (shared with Student A)
Student A's Prediction API	Internal	24/48/72h water level forecasts
Student A's IoT Rainfall Data	Internal	Real-time rainfall intensity
Cloud Database	Common	Shared household data storage
DMC/IOM Historical Data	External	DFRS validation

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ABILITY OF COMMERCIALIZATION

Aspect	Details
Commercialization Potential	High - System addresses critical national need with demonstrated market gap (Cyclone Ditwah 2025)
Target Market	• Primary: Disaster Management Centre (DMC), Irrigation Department • Secondary: Insurance companies for risk assessment • Tertiary: Other river basins in Sri Lanka and South Asia
Customer Profile	Government agencies, NGOs working in disaster management, community-based organizations
Value Proposition	• Reduce flood-related casualties through personalized warnings • Enable efficient resource allocation for evacuations • Ensure inclusive warnings in all national languages • Low-cost, scalable architecture suitable for developing nations
Intellectual Property	• Software copyright for DFRS algorithm • Potential patent for hybrid Bi-LSTM + IoT rainfall integration method • Trademark for system name/brand

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INVESTMENT/COST & RECOVERY

Cost Category	Estimated Cost (LKR)	Details
Labour (Student Research)	Not applicable	Academic project
IoT Hardware (3-5 nodes)	75,000 - 125,000	ESP32, sensors, solar panels, batteries, enclosures
Cloud Services (6 months)	30,000 - 50,000	Database hosting, API endpoints
Mobile App Development	20,000 - 40,000	Developer accounts, testing devices
SMS/Voice Gateway	10,000 - 25,000	For pilot testing
TOTAL ESTIMATED COST	135,000 - 240,000	

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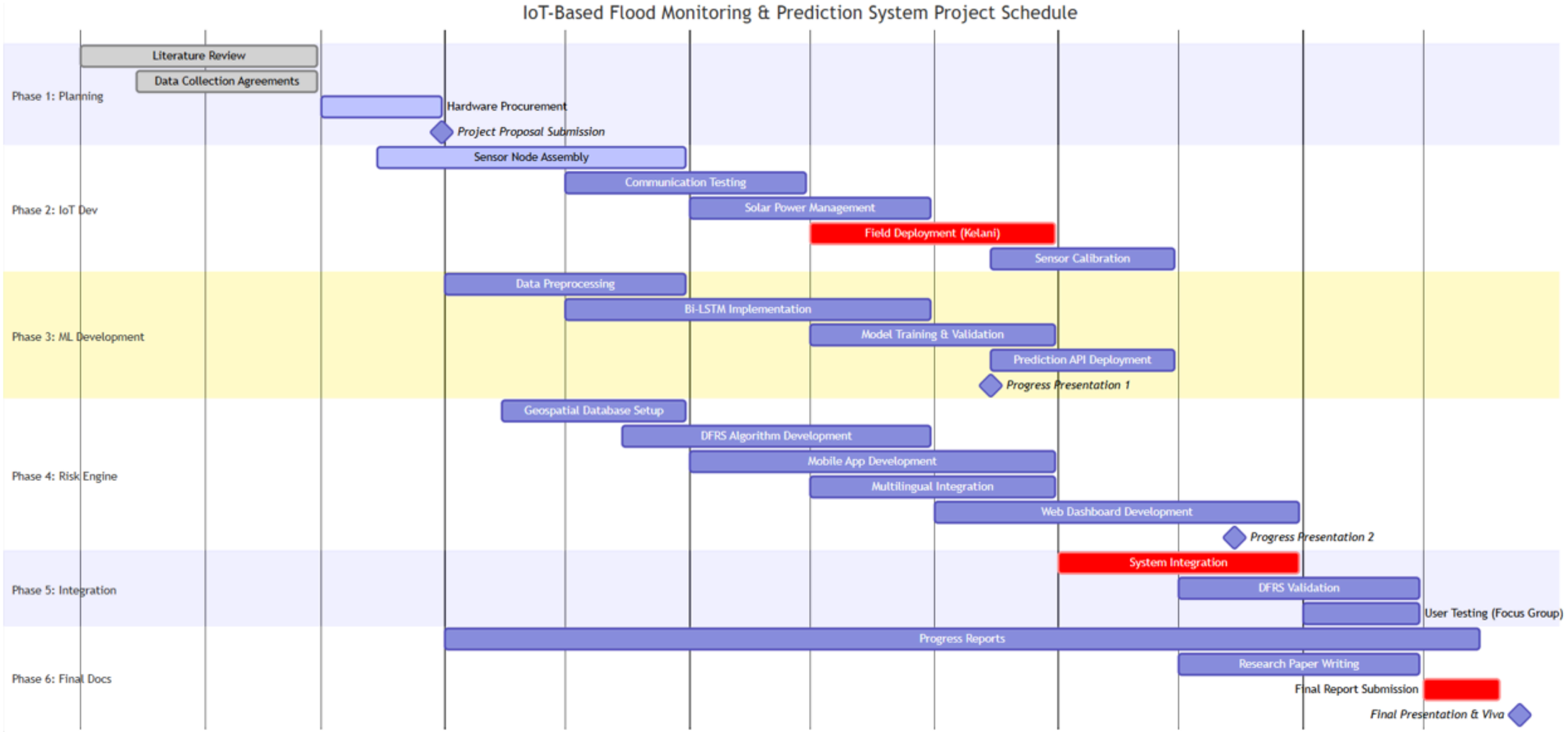

GANTT CHART

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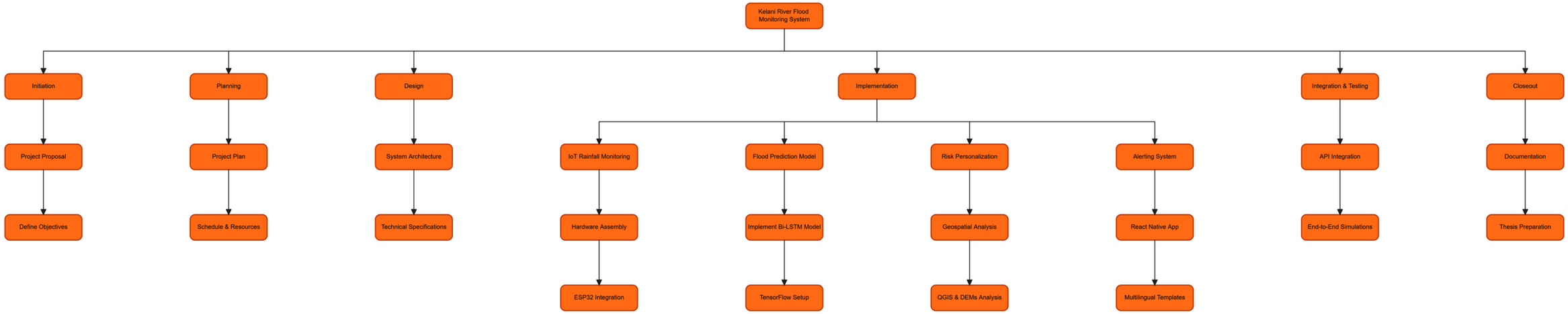
WORK BREAKDOWN STRUCTURE



Thank You